Problem Set 5

Foundations of Machine Learning

- **Submission**: Turn in as a `PDF` and a **separate source code file** on `http://mycourses.unh.edu`. Scanned handwritten notes are encouraged.
- **Questions**: See the class website for a link to office hours

Relevant reading: ILA 6.2, 6.4, 6.5. Problems marked **[code]** should be submitted as a program in a language of your choice (Julia or Python recommended). The cited problem numbers are from ILA, 5th edition (Strang, 2016).

1. ILA Problem Set 6.2 exercises: 1,3,4,6,15, and extra credit: 36,39
2. ILA Problem Set 6.4 exercises: 1,4,6,8,12,24,25,26,34,35
3. ILA Problem Set 6.5 exercises: 2,4,7,14,15,22,23,33
4. Write the following functions in the form $f(x) = x^\top Ax + b^\top x + c$ where $x \in \mathbb{R}^2$ and $A^\top = A$
   (a) $f(x) = x_1^2 + 2x_2^2$
   (b) $f(x) = 3x_1^2 + 2x_2^2 - 2x_1x_2$
   (c) $f(x) = 3x_1^2 + 2x_2^2 - 2x_2x_1 - 2x_1x_1 + 2x_1 - 3$
5. Suppose that $f(x) = x^\top Ax + b^\top x + c$ where $A \in \mathbb{R}^{n\times n}$ may not be symmetric. Show that you can construct a quadratic function $g(x) = x^\top \hat{A}x + \hat{b}^\top x + \hat{c}$ such that $\hat{A}$ is symmetric and $g(x) = f(x), \forall x \in \mathbb{R}^n$.
6. **[code]** Consider a quadratic function $f\colon \mathbb{R}^2 \to \mathbb{R}$ defined as

$$f(x) := \frac{1}{2}\, x^\top \begin{bmatrix} 3 & 1 \\ 1 & 3 \end{bmatrix} x.$$

   Create a contour plot of f and plot the eigenvectors of A onto the plot. Discuss how the eigenvalues relate to the quadratic function.

   Next, consider a quadratic function $g\colon \mathbb{R}^2 \to \mathbb{R}$ defined as

$$g(x) := \frac{1}{2}\, x^\top \begin{bmatrix} 3 & -1 \\ 4 & 3 \end{bmatrix} x,$$

   and repeat the steps above. What do you observe?
7. **Time spent.** Report approximately how many hours you spent completing this problem set (including the reading).